Christopher M. **Ushay, Ph.D.**

56 Lockwood Avenue, Bronxville, NY, USA 10708

+1 914 819 2433 | ushayc@gmail.com | christopherushay.com | christopherushay

Summary

Princeton-trained PhD with experience spanning early-stage startups, award-winning teaching, cross-functional team leadership, and scientific fluency in advanced materials and applied mechanics. Has driven complex projects from ambiguous questions to validated outcomes and translated findings into narratives for both scientific and non-technical audiences.

Education

Princeton University — *Princeton, NJ*

Ph.D., Chemical & Biological Engineering

- Dissertation - "Leveraging Relationships Between Confined Flows and Deformable Media," under advisor Prof. Pierre-Thomas Brun

William E. Macaulay Honors College, City University of New York — *New York, NY*

Bachelor of Engineering, Chemical Engineering, Mathematics Minor

- Valedictorian, Grove School of Engineering at City College of New York | Summa Cum Laude

Work Experience

Independent Contractor, Technical Instruction & AI Training — *Remote*

Self-Employed — *January 2024 - Present*

- Engaged as a domain expert in applied physics, materials science, and engineering to develop and evaluate technical AI training content for frontier research companies
- Produced precise, well-reasoned prompts and responses requiring both scientific fluency and clear written communication

Doctoral Researcher — *Princeton, NJ*

Lab of Pierre-Thomas Brun, Department of Chemical & Biological Engineering, Princeton University — *January 2019 - May 2024*

- Owned multi-year research program end-to-end by scoping open problems in applied physics, setting roadmaps, coordinating collaborators, and delivering validated findings against hard deadlines for publication and conferences
- Transformed technically dense, ambiguous research into a dissertation exploring soft matter and fluid mechanics by distilling research findings into written narratives, including multiple peer-reviewed articles in high-impact journals and a cover page feature
- Wrote and automated software pipelines in Python and Mathematica to convert raw experimental data into quantitative measurements, analyze datasets against theoretical predictions, and visualize relationships to glean publication-grade insights
- Developed 2D and 3D multiphysics simulations of soft material and fluid-structure systems, leveraging models to validate experimental observations and explore design space
- Leveraged machine learning tools to aid in research workflows, including algorithm building, image recognition, scientific ideation, and deep research
- Communicated research findings to global scientific audiences through nearly 10 presentations at major physics conferences and multi-lab collaborations, supported by original figures and multimedia produced in Adobe CC, AutoCAD, and Keynote
- Contributed to scientific art pieces, movies, and sculptures inspired by the lab's research

Assistant in Instruction — *Princeton, NJ*

School of Engineering & Applied Science, Princeton University — *January 2020 - December 2022*

- Recognized with Princeton University's annual Excellence in Teaching Award on student nomination for the instruction of Undergraduate Differential Equations (Spring '20, '21) and Graduate Soft Matter Mechanics (Fall '22)
- Taught intermediate to advanced topics in math and physics to classes of 70+ undergraduate students, spanning classical theory and modern computational tools, by leading weekly lectures, office hours, and ongoing availability as a dedicated resource

Head Manager — *Princeton, NJ*

Debasement Bar, Student-Run Hospitality Venture, Princeton University — *July 2021 - December 2023*

- Managed full P&L operations of a hospitality business, overseeing a 20+ person team, regulatory compliance, nightly event planning, and financial reporting with revenue on the order of tens of thousands
- Maintained facilities (equipment, consumables) and developed data-driven longevity plans with co-managers to ensure continued improvement beyond tenures, including programs such as the implementation of ride-share vouchers for patrons

Machine Learning Research Assistant — *New York, NY*

Element, Inc. — *October 2017 - August 2018*

- Contributed to company's $12M Series A growth phase with cross-functional exposure across Operations and Engineering, including hiring, onboarding, and company culture development, during a period of rapid headcount and process change
- Built and ran data pipelines to process and filter biometric data to support an ML identification platform targeting underserved markets in developing economies

Skills & Interests

- **Languages** - English (fluent); French (CEFR C1); Spanish (CEFR C1); German (CEFR A2); Japanese (elementary)
- **Software & Data Science** - Python (inc. NumPy, pandas, Matplotlib, PyTorch); Mathematica; SQL; COMSOL; MATLAB; C; Julia
- **Design & Visualization** - Adobe CC (Illustrator, Photoshop); Keynote/PowerPoint; AutoCAD; Fiji
- **Fabrication & Analysis** - Laser cutting; CNC milling; 3D printing; digital photography; high-speed imaging; microscopy

Publications

- **Ushay, C.**, Jambon-Puillet, E., Brun, P.-T. (2023) "Interfacial flows through arrays of elastic fibers" Physical Review Fluids, DOI: 10.1103/PhysRevFluids.8.044001
- Badaoui, M., Kresge, G., **Ushay, C.**, Marthelot, J., Brun, P.-T. (2022) "Formation of Pixelated Elastic Films via Capillary Suction of Curable Elastomers in Templated Hele–Shaw Cells" Advanced Materials, DOI: 10.1002/adma.202109682
- Travlou, N.A., **Ushay, C.**, Seredych, M., Rodriguez-Castellon, E., Bandosz, T.J. (2016) "Nitrogen-Doped Activated Carbon-Based Ammonia Sensors: Effect of Specific Surface Functional Groups on Carbon Electronic Properties" ACS Sensors, DOI: 10.1021/acssensors.6b00093

Leadership & Honors

2026	**Board Member**, Princeton University Graduate Alumni Association	*Remote*
2022	**Cover Page Feature**, Advanced Materials, Volume 34, Issue 27	*Print & Online*
2021	**Excellence in Teaching Award**, Princeton University	*Princeton, NJ*
2017	**Valedictorian**, Grove School of Engineering, City College of New York	*New York, NY*
2017	**Alumni Medal for Highest Departmental GPA**, Grove School of Engineering, City College of New York	*New York, NY*

Presentations, Exhibitions, & Press

2024-25	**Rope Piece (Deluge); Deluge 2; Deluge 3**, Traveling Gallery of Fluid Motion: Spiraling Upwards	*Salt Lake City, UT* ***Sculpture***
2023	**"Elastocapillary painting: deposition of fluid trapped within fibre arrays"**, American Physical Society March Meeting 2023	*Las Vegas, NV* ***Presentation***
2022	**"Hairy fluid mechanics: Dynamic elastocapillarity in deformable beam arrays"**, 75th Annual Meeting of the Division of Fluid Dynamics	*Indianapolis, IN* ***Presentation***
2022	**"Fluid mediated assembly of pixelated materials"**, 75th Annual Meeting of the Division of Fluid Dynamics	*Indianapolis, IN* ***Presentation***
2022	**"Hairy fluid mechanics: interfacial flows through networks of flexible fibres"**, 19th U.S. National Congress on Theoretical and Applied Mechanics	*Austin, TX* ***Presentation***
2022	**"Easy as an inkjet, a new soft printing technique has opened the way for pixelated elastics"**, Princeton University Communications	*Online* ***Article***
2021	**"Hairy fluid mechanics: interfacial flows in networks of flexible fibres"**, 74th Annual Meeting of the Division of Fluid Dynamics	*Phoenix, AZ* ***Presentation***
2020	**"Fluid Capture Patterns in Hairy Surfaces"**, 73rd Annual Meeting of the Division of Fluid Dynamics	*Online* ***Presentation***
2020	**"Multiphase Flow Through Hairy Channels"**, American Physical Society March Meeting 2020	*Online* ***Presentation***